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Critical Gaps in Artificial Intelligence in the East and Horn of Africa: A Call to Action to Safeguard Human Rights

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Introduction

Artificial intelligence (AI) is rapidly transforming economies, governance, and social life with the potential to advance sustainable development in Africa (Mienye, Sun, & Ileberi, 2024). The African Union AI strategy posits that AI adoption on the continent will solve the urgent challenges in healthcare, agriculture, education, finance, and public service delivery (African Union [AU], 2024). While AI adoption presents opportunities, it also raises significant concerns, more specifically that most African countries have gaps in regulations of ethical and responsible use of AI (Collaboration on International ICT Policy for East and Southern Africa [CIPESA], 2024).

This article provides an evaluation of the pre-existing AI inequalities and critical gaps in the East and Horn of Africa, which deepen the existing inequalities, facilitate authoritarian control, and undermine democratic governance and human rights. It calls for a proactive and rights-based approach to AI governance that is inclusive, transparent, and contextually relevant.





[Image source: HRRC Canva]

Pre-existing AI inequalities in the East and Horn of Africa

Overall, the AI readiness index 2024 placed Rwanda among the front-runners in being ready for AI. It highlighted countries with notable progress, like Kenya and Ethiopia, and mentioned countries that are lagging, like Eritrea, the Democratic Republic of Congo, Burundi, and South Sudan (CIPESA, 2024).

Pre-existing AI inequalities that led to these determinations include legal issues, regulatory frameworks, and enforcement. Countries in the East and Horn of Africa are at various stages of developing their regulatory frameworks to promote inclusive and responsible AI use. Ethiopia, Ghana, Kenya, and Rwanda have developed AI strategies. Tanzania is in the process of developing its national AI strategies and policies. Somalia and South Sudan have not started the process of developing AI strategies and policies (Research ICT Africa, 2025). Regulatory institutions do not have enough technical capacity and funding to identify, mitigate, or offer redress to AI risks suffered in society (Lawyers Hub, 2025). Weak institutional frameworks, limited judicial capacity, a lack of expertise from policymakers, fragmented laws, and poor enforcement mechanisms are also a concern. Countries where laws exist are seldom enforced (Tech Hive Advisory and Center for Law & Innovation, 2025). The countries in the East and Horn of Africa also lack an AI incident reporting mechanism.

Another AI inequality is the digital divide. There are disparities in digital literacy and access to AI technologies deepening existing inequalities between men and women, affecting already unconnected people, limiting opportunities for some populations, and hindering Africa's competitiveness in the global AI landscape (AU, 2024). For example, Smartphone ownership in Kenya was only about 32% of the population as of 2024 (Global System for Mobile Communications Association [GSMA], 2025). This means a significant portion of the population does not have easy access to the internet and the potential services found on the internet.

Several of these nations face digital infrastructure issues. The limited infrastructure in the region includes unreliable power supplies, unstable internet connectivity, and insufficient digital education programs. These issues continue to hinder widespread adoption of AI (Okolo, 2024; AU, 2024). This inequality in the region has further led to over-reliance on external providers for computational power, including cloud computing services, which pose more risks

related to data sovereignty, privacy, vendor lock-in, and security (World Bank 2024b). For example, AI Hub for Sustainable Development estimates that about 75% of the world's supercomputers are hosted in countries in the Global North, with less than 1% hosted in Africa. The continent also accounts for 2% of global data centers, which poses a risk of creating technological dependencies. Additionally, unreliable electricity could create frequent system downtimes during critical processes. These potential downtimes can remove the benefits of certain AI uses, like for medical diagnoses. If these diagnoses are hindered by a lack of power, it could affect their reliability and use for medical predictions (Segun et al, 2025).

Human Rights Implications of AI Inequality

The inequalities in the growth and application of AI regulations between the East and the Horn of Africa are currently resulting in negative impacts on basic human rights, on the grounds of democracy and electoral integrity, the right to privacy, non-discrimination, climate and environmental impacts, along with the lack of accountability and the state of governance.

Democracy and electoral integrity

AI-powered tools, including social media algorithms and automated bots, are increasingly used to spread disinformation, manipulate public opinion, and suppress dissent during electoral processes (Organization for Security and Co-operation in Europe [OSCE], 2022). The lack of transparency in these systems threatens the integrity of democratic institutions. For instance, some governments in the East and Horn of Africa were reported to deploy sophisticated surveillance technologies from international vendors, which have been used to track and target journalists and political activists, effectively shrinking the civic space (Amnesty International, 2024).

Additionally, many states lack regulatory safeguards, digital literacy programs, or transparency rules for platform political advertising and synthetic content. Tanzania and Uganda, despite having laws governing online communication, have not kept pace with AI-specific threats, leaving their electoral systems vulnerable to new forms of manipulation. External influence from AI technologies developed outside Africa may undermine national sovereignty, Pan-Africanism values, and civil liberties. AI-enabled election manipulation and the dissemination of disinformation pose threats to the integrity of democratic processes, as does the unlawful surveillance of citizens that AI can facilitate (AU, 2024).

Right to Privacy (Data Protection and Surveillance)

For the first time, state agencies can conduct mass surveillance of all citizens' communications, and micro-target individuals for in-depth coverage that compiles real-time data from mobile calls, short message service (SMS), internet messaging, global positioning system (GPS) location, and financial transactions (Institute of Development Studies [IDS], 2023).

A critical gap is the near-total absence of comprehensive data protection laws aligned with the realities of AI. While Kenya has the Data Protection Act (DPA), its enforcement against state-level surveillance remains weak. Amnesty International reported that awareness of data protection regulations and privacy rights among the population is moderate in urban areas and extremely low in rural areas. Even the enforcement mechanisms of these regulations are not often pursued by citizens after experiencing any form of data breach due to a lack of

awareness and trust. There is the belief that any action may be either slow or ineffective (Amnesty International, 2025).

Kenya's Data Protection Act falls short in addressing the specific challenges posed by AI systems. The data processing by these systems can lead to significant privacy violations and other rights infringements. For example, repurposed data might be used for activities like surveillance. Predictive policing creates a society where the right to a private life is severely diminished and potentially stifles freedoms such as expression and association (European Parliament, 2024). Kenya and Uganda have implemented video surveillance, tracking devices, software, and cloud storage systems through a partnership with Chinese telecom firm Huawei under Safe City initiatives. These moves are raising concerns about misuse and a lack of transparency (Lawyers Hub, 2025).

Non-Discrimination: Children, Women, and People with Disabilities

As more children go online and enjoy the benefits of digital technology, they risk their safety and well-being by increasing their exposure to illegal or age-inappropriate content, as well as harmful contact with dangerous adults (UNICEF, 2023). The risks are exacerbated by the lack of robust age verification tools that can mitigate children's exposure to harmful content and interactions online. For instance, in Rwanda and Kenya, the rapid rollout of ed-tech and child-facing chatbots has occurred without robust, privacy-preserving age verification tools or specific legal safeguards for children's data. Safeguards for minors require age-appropriate privacy norms, parental/guardian mechanisms, and algorithmic limits.

Machine learning algorithms can discriminate based on race and gender as well. For example, gender-biased algorithms used in loan applications have been reported to deny women financial independence, and natural language processing tools can perpetuate harmful stereotypes (Buolamwini & Gebru, 2018). Without gender-balanced development teams and audited datasets, AI will entrench patriarchal norms and hinder gender equality. Bias in training data and in algorithmic design can reproduce gender inequalities for such things as health recommendations, credit scoring, and policing. There is limited gender-disaggregated auditing of AI systems and insufficient participation of women and gender minorities in design and governance. AI may amplify gender-based inequality and unfairness unless equity is embedded throughout design and governance. Segun et al cites Raji and Sholademi (2024), who highlight that predictive policing algorithms, developed in vastly different socio-legal contexts, can exacerbate racial and socio-economic profiling if deployed without significant adaptation and monitoring.

AI systems often embed biases, creating new barriers for people with disabilities. For example, facial recognition software used for security and public services in airports and banks across Nairobi, Addis Ababa, and Kigali has not been publicly audited for its accuracy on individuals with disabilities like albinism or facial differences, risking their exclusion.

Right of Access to Social Services and Opportunities

Citron and Pasquale (2014) highlighted that automated scoring systems put crucial opportunities on the line, including the ability to obtain loans, work, housing, and insurance, among others. Deploying AI systems introduces legal and ethical challenges, including algorithmic bias, due process concerns, and accountability gaps in automated decision-making (Ayibam, 2025). There is limited public access to information about datasets, decision rules, or

procurement contracts tied to AI systems. Consequently, this opacity undermines citizens' ability to scrutinize and challenge decisions, thus violating fundamental due process rights.

In Tanzania, they have started using AI to transcribe court rulings, yet concerns remain regarding accuracy, data security, and fairness in automated legal decision-making (Lawyers Hub, 2025). Additionally, Oluka, Mugurusi, Obicci, and Awuor (2022) cite fears of unemployment, the loss of accountability, and widespread bias due to the deployment of AI systems, which undermine fairness.

Institutional Accountability and the Governance Vacuum

Identifying and assessing actual and potential adverse human rights impacts of AI systems is one step in achieving human rights due diligence (Business for Social Responsibility [BSR], 2025). However, most countries in the East and Horn of Africa do not have comprehensive AI regulations. There is a gap in the requirements to assess AI risks. This relates to AI products and services that can be anchored in human rights assessment methodology. Uganda, Kenya, Ethiopia, and Rwanda have national AI strategies. Tanzania has a draft AI strategy under discussion. Other countries in the East and Horn of Africa are yet to commence the process (Africa Data Protection, 2025). Only Rwanda, Kenya, and Djibouti have signed the declaration on sustainable and inclusive AI. The report also notes that no African country has signed the Organization for Economic Co-operation and Development (OECD) Framework for the Classification of AI Systems, adopted in 2022, nor the G7 International Code of Conduct for Organizations Developing Advanced AI Systems, established in 2023. The absence of participation in these frameworks limits the continent's influence in defining international norms and standards relating to AI.

Climate and Environmental Impacts

AI systems, particularly those involving large-scale data processing and machine learning models, consume significant amounts of energy, contributing to environmental degradation and increased carbon emissions (World Bank, 2024b). Additionally, with increasing AI comes the need to replace old hardware with newer efficient modern hardware, generating e-waste, which also affects the environment (AU, 2024; World Bank, 2024). In the East and Horn of Africa, Tanzania, Kenya, and Ethiopia are the highest e-waste generation countries in the region. Only Uganda, Tanzania, and Rwanda have an e-waste policy, legislation, or regulation (Kuehr et al, 2024). The high demand for fresh water to cool data centers poses a threat to regions already facing water scarcity (World Bank, 2024).

Conclusion and Proposed Policy Actions

Countries in the East and Horn of Africa are adopting predominantly foreign developed AI systems in agriculture, healthcare, education, and financial services, which most likely bring bias, misalignments with local values, and disparities in data governance. They also amplify privacy and ethical concerns. The following proposals include policy actions that would guarantee rights, fundamental freedoms, and safety for the people in the region.

- (1) Governments must consider enacting and enforcing comprehensive AI governance frameworks, inspired by the African Union's Data Policy Framework that mandates human rights due diligence, transparency, and accountability. Governments should collaborate on harmonized regulations to be able to hold multinational corporations accountable.

(2) National human rights institutions and data protection authorities in the region must be empowered with technical capacity and legal mandates to audit and regulate AI systems. The institutions should be facilitated to conduct human rights impact assessments and propose mitigation plans for all public AI procurement and high-risk private sector AI. This includes those deployed in health, social protection, elections, and identity management systems, among others. Regional Economic Communities should design a regional AI incident reporting platform to be able to counter misinformation and data breaches. It could also hold relevant parties accountable.

(3) Governments should establish or empower independent, well-resourced AI and data governance institutions, including at the regional level. These institutions must have clear mandates for oversight and the capacity to audit AI systems, investigate rights violations, and handle complaints effectively. These institutions should investigate AI trends, analyze rights infringement cases, and consider risk countermeasures. This includes creating a regional platform that facilitates the anonymized and publicly motivated exchange of data such as incident reports, testing outcomes, and known vulnerabilities, which would help support early warning capabilities and enable coordinated responses.

(4) Countries should invest in digital infrastructure and inclusion. These investments should include expanding internet access, reducing device and access costs, and improving digital infrastructure, especially in marginalized and underserved areas. AI strategies and tools should be developed to prioritize local languages and cultural relevance. They should respond to local needs and realities.

(5) Countries should ensure that all AI deployed and used in the public sector is transparent and auditable, including publishing procurement disclosures, providing clear explanations of AI systems' purpose, usage, and implications. All safeguards should be publicly accessible. Vendors supplying AI tools and applications to governments must be required to disclose key details about their technologies, including their data sources, algorithms, and potential biases.

(6) Governments must also adopt policies that limit the use of AI in sensitive domains, such as surveillance and criminal justice, unless robust safeguards are in place to prevent misuse and mitigate harms that may occur in their respective usage.

(7) Governments should strengthen and integrate detection systems for deep fakes, hate speech, and coordinated disinformation campaigns. They should implement clear tagging or watermarking of AI-generated content and ensure timely responses to reports of harmful content from the region.

(8) Governments should build out the technical and research capacity of key stakeholders to monitor and document the impact of AI on human rights (social, economic, and political). This includes establishing accessible reporting channels for AI abuse and collecting evidence-based case studies of AI abuses and their effects, particularly on vulnerable, rural, and marginalized groups.

- **AI Governance:** a system of policies, processes, people, and daily practices that enable the responsible development, deployment, and oversight of AI systems.
- **AI Incident:** an event, circumstance, or series of events where the development, use, or malfunction of one or more AI systems directly or indirectly leads to a specific harm. These harms include injury to health, disruption of critical infrastructure, human rights violations, or damage to property, communities, or the environment.
- **AI safety:** a scientific field concerned with identifying and addressing the security, ethical, socioeconomic, environmental, technical, and existential risks and harms of frontier AI models.
- **Artificial Intelligence (AI):** an AI system is a machine-based system that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments. Different AI systems vary in their levels of autonomy and adaptiveness after deployment.
- **Audits:** the systematic examination of the algorithms and data used in an AI system to assess its fairness, accountability, transparency, and ethical implications.
- **Bias in Artificial Intelligence (AI):** systematic errors in AI systems that lead to unfair or discriminatory outcomes, often favoring or disadvantaging certain groups.
- **Computational Power:** the ability to store, process, and transfer data at scale. This is crucial for training and deploying AI models and applications.
- **Human Rights Impact Assessment (HRIA):** a process for identifying and assessing actual and potential human rights impacts of an AI system as well as addressing them based on guidance in the UN Guiding Principles on Business and Human Rights.

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